

Crushing and Splitting DAAs for HCV Treatment: A Case Series

Kristen Whelchel¹, Autumn D. Zuckerman¹, David E. Koren², Caroline Derrick³, Jeannette Bouchard⁴, Cody Chastain⁵

¹ Specialty Pharmacy Services, Vanderbilt University Medical Center; Nashville, Tennessee ² Temple University Health System; Philadelphia, Pennsylvania
³ University of South Carolina Department of Infectious Disease; Columbia, South Carolina ⁴ WakeMed Health and Hospital System; Raleigh, North Carolina
⁵ Division of Infectious Diseases, Department of Medicine, Vanderbilt University Medical Center; Nashville, Tennessee



726 Melrose Avenue Nashville, TN 37211/Email: kristen.whelchel@vumc.org/Tel: 615.875.6131 Fax: 615.875.0666

BACKGROUND

- Direct Acting Antivirals (DAAs) can produce sustained virologic response (SVR) rates >90%.
- There is limited data regarding the use of DAAs in patients unable to swallow tablets.
- DAA tablet manipulation may impact drug absorption and treatment outcome.

OBJECTIVE

- Describe the safety and effectiveness outcomes of real-world cases requiring DAA tablet manipulation.

METHOD

Design	Multi-site, retrospective case series
Sample	Adult patients receiving DAA therapy with tablet manipulation at three academic health-systems
Study period	January 2013 to December 2019
Outcomes	Achievement of SVR at least 12 weeks after therapy completion, reasons for tablet manipulation, adverse effects and adherence

Figure 1: Practice Sites

Vanderbilt University Medical Center Nashville, TN Infectious Diseases Clinic Physicians 1 Nurse Practitioners 1 Pharmacists 1 CPA	Prisma Health-USC Immunology Center Columbia, SC Infectious Diseases Clinic Physicians 15 Nurse Practitioners 3 Pharmacists 2	Temple University Philadelphia, PA Internal Medicine and Infectious Diseases Clinic Physicians 4 Fellows 6 Nurse Practitioners 2 Pharmacists 1 CPA
--	--	--

CPA, collaborative practice agreement

REFERENCES

1. World Health Organization. Hepatitis C fact sheet. 2019. <https://www.who.int/news-room/fact-sheets/detail/hepatitis-c>. Accessed October 6, 2020.
2. Oberoi RK, Zhao W, Sidhu DS, et al. A phase 1 study to evaluate the effect of crushing, cutting into half, or grinding of glecaprevir/pibrentasvir tablets on exposures in healthy subjects. J Pharm Sci. 2018;107:1724-30.

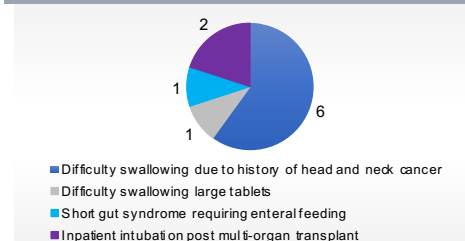
RESULTS

Table 1: Summary of Cases of HCV Treatment Requiring DAA Manipulation

Gender	Race	Age	Pertinent Medical History	Non-DAA RX Burden	GT	Fibrosis Stage	Previous HCV Treatment	Drug Regimen	Potential Drug Interactions with DAA	Method of Administration	Patient-Reported Adherence	Treatment Outcome
Male	White	67	NAT+ heart/kidney transplant, HTN, HLD, DM, GI bleeds	28	1a	Not reported	Naive	GLE/PIB	atorvastatin, quetiapine, tacrolimus, omeprazole	Crushed and taken by PEG tube	No missed doses	SVR12 achieved
Female	Black	68	NAT+ heart/kidney transplant, TTR amyloidosis, ESRD	30	1a	Not reported	Naive	GLE/PIB	pantoprazole, oxycodone, tacrolimus	Crushed and taken by PEG tube	2 missed doses	SVR12 achieved
Male	Black	71	Short gut syndrome, ischemic colitis requiring colectomy	8	1a	F2-F3	Naive	LDV/SOF	N/A	Halved and taken by mouth	No missed doses	SVR12 achieved
Male	Black	61	H/o squamous cell carcinoma of larynx, HTN, DM, HLD	7	1a	F0	Experienced (IFN)	LDV/SOF	magnesium	Crushed and taken with small amount of orange juice	No missed doses	Lost to follow up
Male	Black	54	H/o laryngeal cancer	3	1a	F0-F1	Naive	LDV/SOF	N/A	Crushed and taken by mouth	Several missed doses	Lost to follow up
Male	White	60	H/o carcinoma of tonsil, HCC, GERD, HTN	8	3	F2-F3	Naive	SOF/VEL	N/A	Crushed and taken by PEG tube	1 missed dose	SVR12 achieved
Male	Black	73	H/o malignant neoplasm of supraglottis	3	3	F2	Naive	SOF/VEL	N/A	Crushed and taken by mouth	No missed doses	SVR12 achieved
Female	White	41	Scoliosis w/Harrington rod, BMI 17.8	0	3	F0	Naive	SOF/VEL	N/A	Halved and taken on gelatin	No missed doses	SVR12 achieved
Female	White	60	H/o squamous cell carcinoma of larynx, GERD HTN, CAD	11	1a	F0	Experienced (SMV + SOF)	SOF/VEL	N/A	Crushed and taken sprinkled on applesauce	31 missed doses	SVR12 achieved
Female	White	50	Decompensated cirrhosis, h/o submandibular malignant mass	12	3	F4	Experienced (IFN)	SOF/VEL + RBV	calcium carbonate, ranitidine	Quartered and taken by mouth	4 missed doses	Lost to follow up

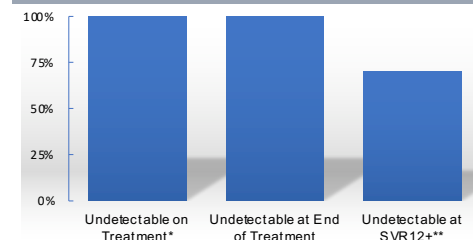
BMI, body mass index; CAD, coronary artery disease; DM, diabetes mellitus; ESRD, end stage renal disease; GERD, gastroesophageal reflux; GI, gastrointestinal; GLE/PIB, glecaprevir/pibrentasvir; GT, genotype; HCC, hepatocellular carcinoma; HLD, hyperlipidemia; H/o, history of; HTN, hypertension; IFN, interferon; LDV/SOF, ledipasvir/sofosbuvir; NAT, nucleic acid testing; PEG, percutaneous endoscopic gastrostomy; RBV, ribavirin; RX, prescription; SMV, simeprevir; SOF, sofosbuvir; SOF/VEL, sofosbuvir/velpatasvir; TTR, transthyretin

Figure 2: Reasons for Tablet Manipulation



- No patients experienced severe adverse effects.
- Unpleasant taste was reported.

Figure 3: HCV RNA Lab Monitoring



*Defined as reaching an undetectable HCV RNA while on treatment
**3 patients lost to follow up after end of treatment

CONCLUSION

- All patients with available data achieved an SVR12.
- This case series provides evidence for safety and effectiveness with HCV DAA tablet manipulation.

DISCLOSURES

- David E. Koren is an independent consultant for AbbVie and has participated on an advisory panel for Gilead.
- Other authors report no disclosures or conflicts of interests.

NASP ANNUAL MEETING, SEPTEMBER 27-30, 2021